

Questions

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Q1 - 2024 (04 Apr Shift 1)

Three urns A, B and C contain 7 red, 5 black; 5 red, 7 black and 6 red, 6 black balls, respectively. One of the urn is selected at random and a ball is drawn from it. If the ball drawn is black, then the probability that it is drawn from urn A is :

- (1) $\frac{5}{18}$
- (2) $\frac{5}{16}$
- (3) $\frac{4}{17}$
- (4) $\frac{7}{18}$

Q2 - 2024 (04 Apr Shift 2)

In a tournament, a team plays 10 matches with probabilities of winning and losing each match as $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Let x be the number of matches that the team wins, and y be the number of matches that team loses. If the probability $P(|x - y| \leq 2)$ is p , then $3^9 p$ equals _____

Q3 - 2024 (05 Apr Shift 1)

The coefficients a, b, c in the quadratic equation $ax^2 + bx + c = 0$ are chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. The probability of this equation having repeated roots is :

- (1) $\frac{1}{128}$
- (2) $\frac{1}{64}$
- (3) $\frac{3}{256}$
- (4) $\frac{3}{128}$

Q4 - 2024 (05 Apr Shift 2)

The coefficients a, b, c in the quadratic equation $ax^2 + bx + c = 0$ are from the set $\{1, 2, 3, 4, 5, 6\}$. If the probability of this equation having one real root bigger than the other is p , then $216 p$ equals :

- (1) 57
- (2) 76

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(3) 38

(4) 19

Q5 - 2024 (05 Apr Shift 2)

Let the mean and the standard deviation of the probability distribution

X	α	1	0	-3
P(X)	$\frac{1}{3}$	K	$\frac{1}{6}$	$\frac{1}{4}$

be μ and σ , respectively. If $\sigma - \mu = 2$, then $\sigma + \mu$ is equal to _____

Q6 - 2024 (06 Apr Shift 1)

A company has two plants A and B to manufacture motorcycles. 60% motorcycles are manufactured at plant A and the remaining are manufactured at plant B. 80% of the motorcycles manufactured at plant A are rated of the standard quality, while 90% of the motorcycles manufactured at plant B are rated of the standard quality. A motorcycle picked up randomly from the total production is found to be of the standard quality. If p is the probability that it was manufactured at plant B, then $126p$ is

(1) 54

(2) 66

(3) 64

(4) 56

Q7 - 2024 (06 Apr Shift 2)

If three letters can be posted to any one of the 5 different addresses, then the probability that the three letters are posted to exactly two addresses is:

(1) $\frac{18}{25}$ (2) $\frac{12}{25}$ (3) $\frac{6}{25}$ (4) $\frac{4}{25}$

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Q8 - 2024 (06 Apr Shift 2)

From a lot of 12 items containing 3 defectives, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Let items in the sample be drawn one by one without replacement. If variance of X is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $n - m$ is equal to _____

Q9 - 2024 (08 Apr Shift 1)

Let the sum of two positive integers be 24. If the probability, that their product is not less than $\frac{3}{4}$ times their greatest possible product, is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $n - m$ equals

- (1) 10
- (2) 9
- (3) 11
- (4) 8

Q10 - 2024 (08 Apr Shift 1)

Three balls are drawn at random from a bag containing 5 blue and 4 yellow balls. Let the random variables X and Y respectively denote the number of blue and yellow balls. If $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively, then $7\bar{X} + 4\bar{Y}$ is equal to _____

Q11 - 2024 (08 Apr Shift 2)

There are three bags X , Y and Z . Bag X contains 5 one-rupee coins and 4 five-rupee coins; Bag Y contains 4 one-rupee coins and 5 five-rupee coins and Bag Z contains 3 one-rupee coins and 6 five-rupee coins. A bag is selected at random and a coin drawn from it at random is found to be a one-rupee coin. Then the probability, that it came from bag Y , is :

- (1) $\frac{1}{4}$
- (2) $\frac{1}{2}$
- (3) $\frac{5}{12}$
- (4) $\frac{1}{3}$

Questions

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Q12 - 2024 (09 Apr Shift 1)

Let a , b and c denote the outcome of three independent rolls of a fair tetrahedral die, whose four faces are marked 1, 2, 3, 4. If the probability that $ax^2 + bx + c = 0$ has all real roots is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____

Q13 - 2024 (09 Apr Shift 2)

If an unbiased dice is rolled thrice, then the probability of getting a greater number in the i^{th} roll than the number obtained in the $(i - 1)^{\text{th}}$ roll, $i = 2, 3$, is equal to

(1) $\frac{3}{54}$

(2) $\frac{2}{54}$

(3) $\frac{1}{54}$

(4) $\frac{5}{54}$

Questions

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Answer Key

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Q5 (5) MathonGo Q6 (1) MathonGo Q7 (2) MathonGo Q8 (71) MathonGo

Q9 (1) MathonGo Q10 (17) MathonGo Q11 (4) MathonGo Q12 (19) MathonGo

Q13 (4) MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

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Solutions

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Q1

$$\begin{array}{c}
 \text{A} \quad \text{B} \quad \text{C} \\
 7R, 5B \quad 5R, 7B \quad 6R, 6B \\
 P(B) = \frac{1}{3} \cdot \frac{5}{12} + \frac{1}{3} \cdot \frac{7}{12} + \frac{1}{3} \cdot \frac{6}{12} \\
 \text{required probability} = \frac{\frac{1}{3} \cdot \frac{5}{12}}{\frac{1}{3} \cdot \left[\frac{5}{12} + \frac{7}{12} + \frac{6}{12} \right]} = \frac{5}{18}
 \end{array}$$

Q2

$$P(W) = \frac{1}{3} \quad P(L) = \frac{2}{3}$$

 x = number of matches that team wins y = number of matches that team loses

$$|x - y| \leq 2 \text{ and } x + y = 10$$

$$|x - y| = 0, 1, 2 \quad x, y \in N$$

Case-I

$$- I : |x - y| = 0 \Rightarrow x = y$$

$$\therefore x + y = 10 \Rightarrow x = 5 = y$$

$$P(|x - y| = 0) = {}^{10}C_5 \left(\frac{1}{3} \right)^5 \left(\frac{2}{3} \right)^5$$

$$\text{Case-II : } |x - y| = 1 \Rightarrow x - y = \pm 1$$

$x = y + 1$	$x = y - 1$
$\therefore x + y = 10$	$\therefore x + y = 10$
$2y = 9$	$2y = 11$
Not possible	Not possible

$$\text{Case-III : } |x - y| = 2 \Rightarrow x - y = \pm 2$$

$$x - y = 2 \quad \text{OR} \quad x - y = -2$$

$$\therefore x + y = 10 \quad \therefore x + y = 10$$

$$x = 6, y = 4 \quad x = 4, y = 6$$

$$P(|x - y| = 2) = {}^{10}C_6 \left(\frac{1}{3} \right)^6 \left(\frac{2}{3} \right)^4 + {}^{10}C_4 \left(\frac{1}{3} \right)^4 \left(\frac{2}{3} \right)^6$$

$$p = {}^{10}C_5 \frac{2^5}{3^{10}} + {}^{10}C_6 \frac{2^4}{3^{10}} + {}^{10}C_4 \frac{2^6}{3^{10}}$$

$$\begin{aligned}
 3^9 p &= \frac{1}{3} ({}^{10}C_5 2^5 + {}^{10}C_6 2^4 + {}^{10}C_4 2^6) \\
 &= 8288
 \end{aligned}$$

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Solutions

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Q3

$$ax^2 + bx + c = 0$$

$$a, b, c \in \{1, 2, 3, 4, 5, 6, 7, 8\}$$

Repeated roots $D = 0$

$$\Rightarrow b^2 - 4ac = 0 \Rightarrow b^2 = 4ac$$

$$\text{Prob} = \frac{8}{8 \times 8 \times 8} = \frac{1}{64}$$

$$\Rightarrow (a, b, c)$$

$$(1, 2, 1); (2, 4, 2); (1, 4, 4); (4, 4, 1); (3, 6, 3);$$

$$(2, 8, 8); (8, 8, 2); (4, 8, 4)$$

8 case

Q4

$$D > 0$$

$$b^2 > 4ac$$

$$b = 3 : (a, c) = (1, 1)(1, 2)(2, 1)$$

$$b = 4 : (a, c) = (1, 1)(1, 2)(2, 1)(1, 3)(3, 1)$$

$$b = 5 : (a, c) = (1, 1)(1, 2)(2, 1)(1, 3)(3, 1)(1, 4)(4, 1)$$

$$(1, 5)(5, 1)(1, 6)(6, 1)(2, 3)(3, 2)(2, 2)$$

$$b = 6 : (a, c) = (1, 1)(1, 2)(2, 1)(1, 3)(3, 1)(1, 4)(4, 1)$$

$$(1, 5)(5, 1)(1, 6)(6, 1)(2, 3)(3, 2)(2, 4)(4, 2)(2, 2)$$

fav. cases = 38

$$\text{Prob.} : \frac{38}{6 \times 6 \times 6}$$

Q5

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Solutions

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$$\frac{1}{3} + k + \frac{1}{6} + \frac{1}{4} = 1 \Rightarrow k = \frac{1}{4}$$

$$\mu = \frac{\alpha}{3} + \frac{1}{4} - \frac{3}{4}$$

$$\mu = \frac{\alpha}{3} - \frac{1}{2}$$

$$\sigma = \sqrt{\left(\alpha^2 \frac{1}{3} + \frac{1}{4} + 9 \frac{1}{4}\right) - \left(\frac{\alpha}{3} - \frac{1}{2}\right)^2}$$

$$\sigma = \sqrt{\frac{2\alpha^2}{9} + \frac{\alpha}{3} + \frac{9}{4}}$$

$$\sigma = \mu + 2$$

$$\sigma^2 = (\mu + 2)^2 \Rightarrow \frac{2\alpha^2}{9} + \frac{\alpha}{3} + \frac{9}{4} = \frac{\alpha^2}{9} + \frac{9}{4} + \alpha$$

$$\frac{\alpha^2}{9} - \frac{2\alpha}{3} = 0$$

$$\alpha = 0, (\text{reject}) \text{ or } \alpha = 6$$

$$(\because x = 0 \text{ is already given})$$

$$\Rightarrow \sigma + \mu = 2\mu + 2$$

$$= 5$$

Q6

	A	B
Manufactured	60%	40%
Standard quality	80%	90%

P (Manufactured at B/ found standard quality) = ?

A : Found S.Q

B : Manufacture B

C : Manufacture A

$$P(E_1) = \frac{40}{100}$$

$$P(E_2) = \frac{60}{100}$$

$$P(A/E_1) = \frac{90}{100}$$

$$P(A/E_2) = \frac{80}{100}$$

$$\therefore P(E_1/A) = \frac{P(A/E_1)P(E_1)}{P(A/E_1)P(E_1) + P(A/E_2)P(E_2)} = \frac{3}{7}$$

$$\therefore 126P = 54$$

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Solutions

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Q7

$$\begin{aligned} \text{Total method} &= 5^3 \\ \text{favorable} &= {}^5C_2 (2^3 - 2) = 60 \end{aligned}$$

$$\text{probability} = \frac{60}{125} = \frac{12}{25}$$

Q8

$$a = 1 - \frac{{}^3C_5}{{}^{12}C_5}$$

$$b = 3 \cdot \frac{{}^9C_4}{{}^{12}C_5}$$

$$c = 3 \cdot \frac{{}^9C_3}{{}^{12}C_5}$$

$$d = 1 \cdot \frac{{}^9C_2}{{}^{12}C_5}$$

$$u = 0 \cdot a + 1 \cdot b + 2 \cdot c + 3 \cdot d = 1.25$$

$$\sigma^2 = 0 \cdot a + 1 \cdot b + 4 \cdot c + 9 \cdot d - u^2$$

$$\sigma^2 = \frac{105}{176}$$

$$\text{Ans. } 176 - 105 = 71$$

Q9

$$x + y = 24, x, y \in \mathbb{N}$$

$$AM > GM \Rightarrow xy < 144$$

$$xy \geq 108$$

Favorable pairs of (x, y) are

$$(13, 11), (12, 12), (14, 10), (15, 9), (16, 8),$$

$$(17, 7), (18, 6), (6, 18), (7, 17), (8, 16), (9, 15),$$

$$(10, 14), (11, 13)$$

i.e. 13 cases

$$\text{Total choices for } x + y = 24 \text{ is } 23$$

$$\text{Probability} = \frac{13}{23} = \frac{m}{n}$$

$$n - m = 10$$

Q10

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Solutions

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Blue balls	0	1	2	3	4	5
Prob.	$\frac{{}^5C_0 \cdot {}^4C_1}{{}^9C_3}$	$\frac{{}^5C_1 \cdot {}^4C_2}{{}^9C_3}$	$\frac{{}^5C_2 \cdot {}^4C_1}{{}^9C_3}$	$\frac{{}^5C_3 \cdot {}^4C_0}{{}^9C_3}$	0	0

$$7\bar{x} = \frac{{}^5C_1 \cdot {}^4C_2 + {}^5C_2 \cdot {}^4C_1 \times 2 + {}^5C_3 \cdot {}^4C_0 \times 3}{{}^9C_3} \times 7$$

$$\frac{30 + 80 + 30}{84} \times 7$$

$$= \frac{140}{12} = \frac{70}{6} = \frac{35}{3}$$

yellow	0	1	2	3	4
		${}^5C_2 \cdot {}^4C_1$	${}^5C_1 \cdot {}^4C_2$	${}^5C_0 \cdot {}^4C_3$	0

$$4\bar{y} = \frac{40+60+12}{84} \times 4 = \frac{112}{21} = \frac{16}{3}$$

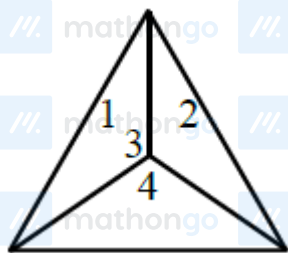
Q11

X Y Z
 5 one & 4 five 4 one & 5 five 3 one & 6 five

$$P = \frac{4/9}{5/9 + 4/9 + 3/9} = \frac{4}{12} = \frac{1}{3}$$

Q12

$$a, b, c \in \{1, 2, 3, 4\}$$



Tetrahedral dice

$$ax^2 + bx + c = 0$$

has all real roots

$$\Rightarrow D \geq 0$$

$$\Rightarrow b^2 - 4ac \geq 0$$

$$\text{Let } b = 1 \Rightarrow 1 - 4ac \geq 0 \text{ (Not feasible)}$$

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Solutions

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$$b = 2 \Rightarrow 4 - 4ac \geq 0$$

$$1 \geq ac \Rightarrow a = 1, c = 1,$$

$$b = 3 \Rightarrow 9 - 4ac \geq 0$$

$$\frac{9}{4} \geq ac$$

$$\Rightarrow a = 1, c = 1$$

$$\Rightarrow a = 1, c = 2$$

$$\Rightarrow a = 2, c = 1$$

$$b = 4 \Rightarrow 16 - 4ac \geq 0$$

$$4 \geq ac$$

$$\Rightarrow a = 1, c = 1$$

$$\Rightarrow a = 1, c = 2 \quad \Rightarrow a = 2, c = 1$$

$$\Rightarrow a = 1, c = 3 \quad \Rightarrow a = 3, c = 1$$

$$\Rightarrow a = 1, c = 4 \quad \Rightarrow a = 4, c = 1$$

$$\Rightarrow a = 2, c = 2$$

$$\text{Probability} = \frac{12}{(4)(4)(4)} = \frac{3}{16} = \frac{m}{m}$$

$$m + n = 19$$

Q13

$$\text{Favourable cases} = {}^6C_3$$

$$\text{Total out comes} = 6^3$$

Probability of getting greater number than previous

$$\text{one} = \frac{{}^6C_3}{r^3} = \frac{20}{216} = \frac{5}{54}$$

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